Project Report : CI/CD Scientific Calculator

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Project: Scientific Calculator CI/CD Pipeline

Technologies: Java, Maven, Jenkins, Docker, Ansible

1. What and Why of DevOps

DevOps is a professional movement that emphasizes communication, collaboration, and integration between software developers (Dev) and IT operations (Ops). It aims to automate the process of software delivery and infrastructure changes.

DevOps enables faster release cycles, improved reliability, and greater efficiency. In this project, it ensures that every change to the scientific calculator is automatically tested and deployed without manual intervention.

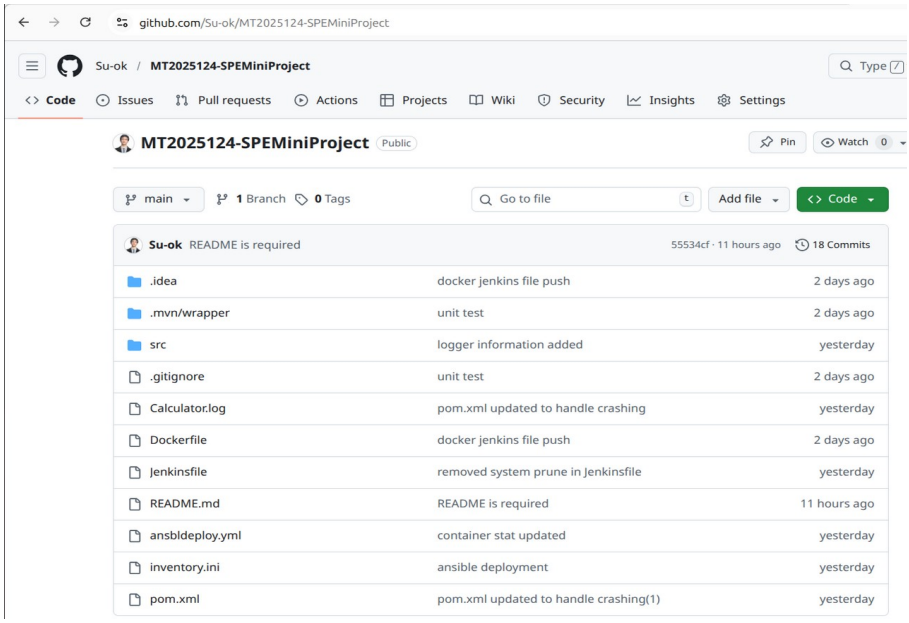
2. Tools Used

GitHub – Source code version control
Apache Maven – Build automation and dependency management
JUnit 4 – Unit testing framework
Jenkins – Continuous Integration pipeline
Docker – Containerization platform
Ansible – Deployment automation

3. Step-by-Step Implementation

Step 1: Source Control Management (GitHub)

Code is managed in a GitHub repository to track changes and trigger automated builds.

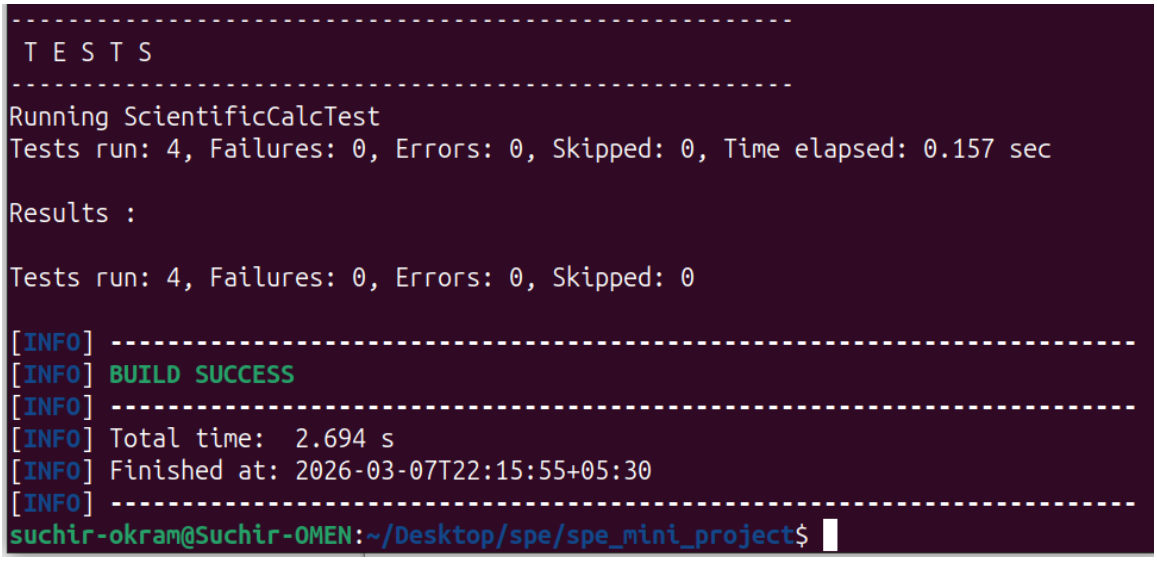


Step 2: Build and Testing (Maven & JUnit)

Maven compiles the project and manages dependencies while JUnit verifies the mathematical logic.

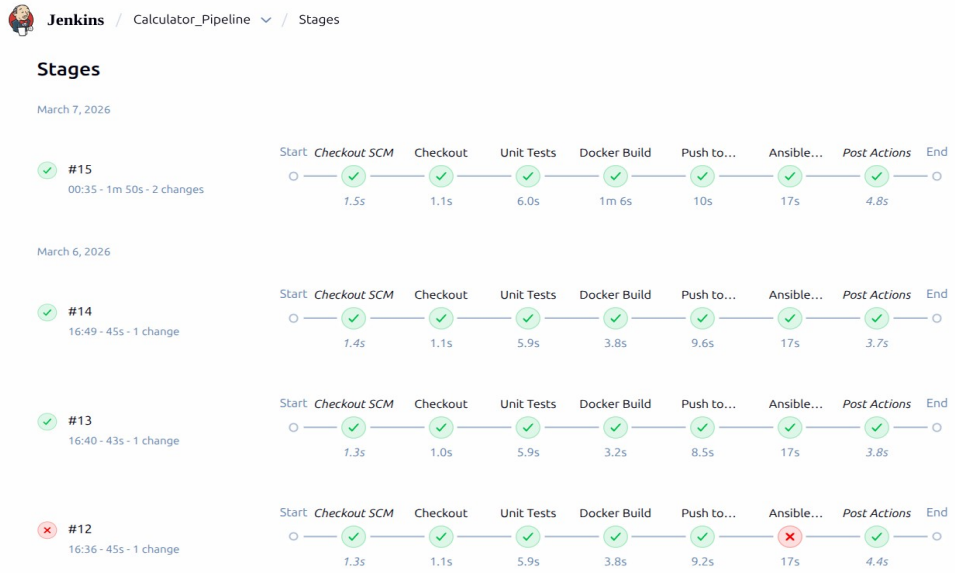
Commands:

```
mvn clean compile
mvn test
```



Step 3: Continuous Integration (Jenkins)

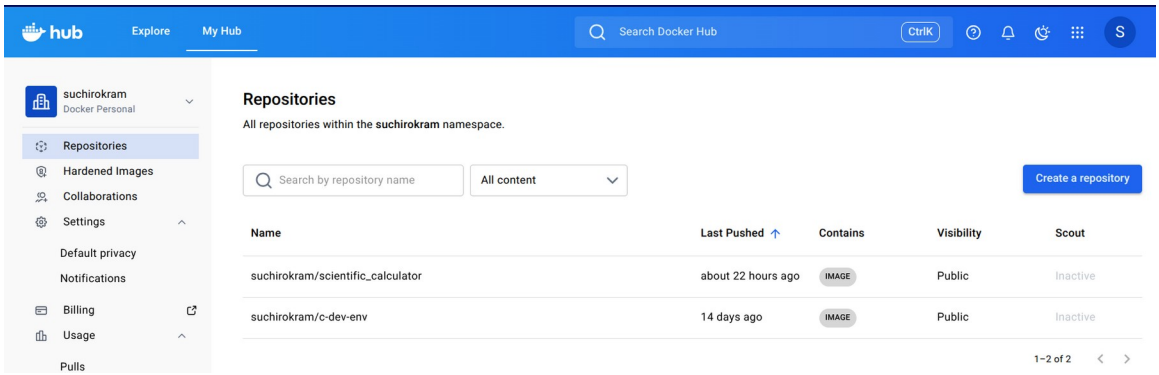
Jenkins automates building, testing, and deployment whenever code is pushed to GitHub.



Step 4: Containerization (Docker)

The application is packaged inside a Docker container so that it runs consistently across environments.

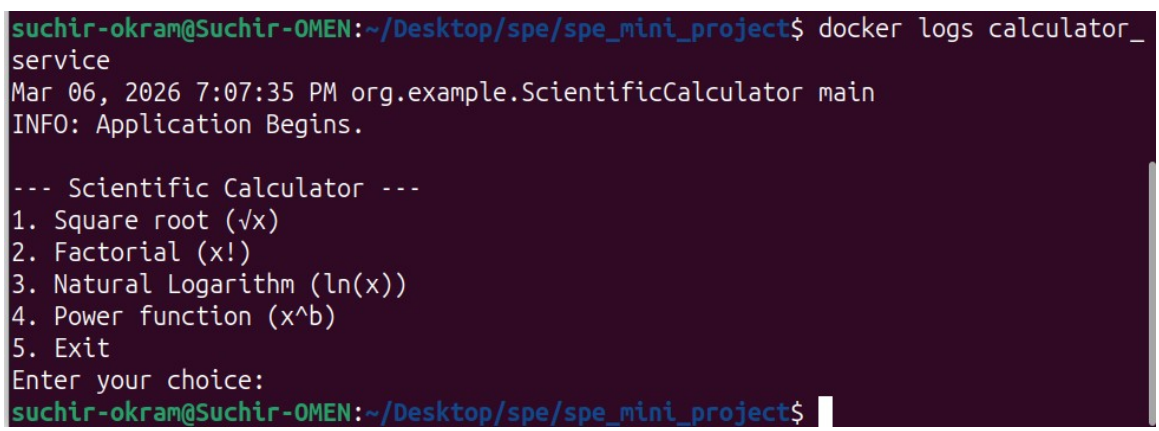
Command: `docker build -t suchirokram/scientific_calculator:latest .`

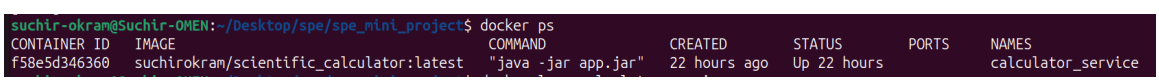


Step 5: Deployment (Ansible)

Ansible automates deployment by pulling the latest Docker image and starting the container.

Command: `ansible-playbook -i inventory.ini ansbldeploy.yml`





4. Application Verification

Run the calculator inside the container using:

```
docker exec -it calculator_service java -jar app.jar
```

Screenshots showing calculations for:

• Square Root

```
Enter your choice: 1
Enter number: 25
Result: 5.0
Mar 07, 2026 5:26:37 PM org.example.ScientificCalculator main
INFO: Square Root of 25.0 = 5.0
```

• Factorial

```
Enter your choice: 2
Enter integer: 7
Result: 5040
Mar 07, 2026 5:36:10 PM org.example.ScientificCalculator main
INFO: Factorial of 7 = 5040
```

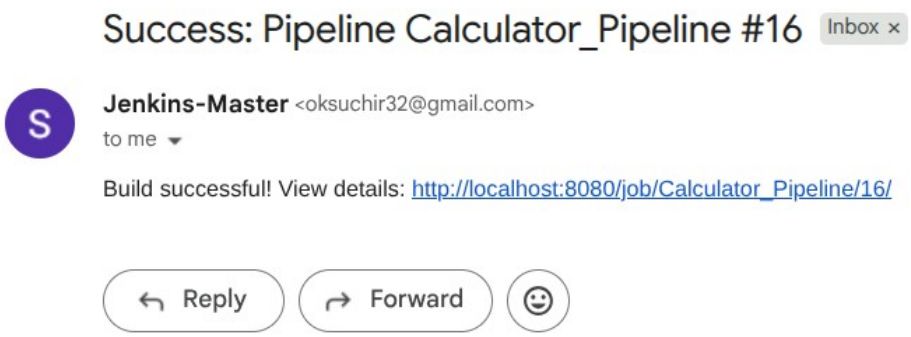
• Natural Logarithm

```
Enter your choice: 3
Enter number: 2.8
Result: 1.0296194171811581
Mar 07, 2026 5:27:56 PM org.example.ScientificCalculator main
INFO: Natural Log of 2.8 = 1.0296194171811581
```

• Power Function

```
Enter your choice: 4
Enter base: 9
Enter exponent: 3
Result: 729.0
Mar 07, 2026 5:28:13 PM org.example.ScientificCalculator main
INFO: Power 9.0^3.0 = 729.0
```

Email Notification:



5. Phases of the CI/CD Pipeline

Phase	Tool	Key Responsibility
Trigger	GitHub + **ngrok**	Detects code changes and bypasses local firewall.
CI	Jenkins + Maven	Compiles and validates logic via JUnit.
Storage	Docker Hub	Hosts the verified, ready-to-use application image.
CD	Ansible	Deploys the container in the correct "Interactive" state.
Alerting	Jenkins Mail	Notifies **noiropx59@gmail.com** of the final status.

6. Repository Links

GitHub: https://github.com/Su-ok/MT2025124-SPEMiniProject

Docker Hub: https://hub.docker.com/r/suchirokram/scientific_calculator/tags

7. Project Files (Appendix)

Dockerfile – Multi-stage Docker build configuration.

Jenkinsfile – Defines CI/CD pipeline stages.

ansbldeploy.yml – Ansible deployment playbook.